Jane Im's Research Statement

My research in **Human-Computer Interaction** and **Social Computing** develops **consentful systems**, which re-envision how users of social technologies grant consent in order to tackle issues around privacy and safety, especially in the context of social media. My key insight is that consent is a better framework for considering a range of privacy, permissions, and other related issues, because traditional permission models tend to be system and outcome-centric, whereas consent is inherently user-centric and process-focused. For example, the best users can do with privacy settings is block or mute people, or set a post to be private or shown to only "friends." These are engineers' or a lawyers' conception of privacy, where ease of programming or fidelity to some policy might be the primary goal. But, the forms of information exchange that people consent to are dependent on identities or relationships, and privacy settings do not come close to representing such complexity. The stakes are high when sensitive information is involved, such as in reporting systems for escalating abuse of power. In these cases, failure to provide flexible consent-granting mechanisms can have serious consequences for individuals, which prevents people from reporting in the first place.

A consent focus would not be satisfied until users can adjust who they interact with, clearly, easily, flexibly, and at any time. However, there has been a lack of a clear definition of consent that can be applied to systems across domains, as well as insights on how to exactly embed consent into systems. In my research, I:

- **theorize what affirmative consent is**—affirmative consent is *voluntary*, *informed*, *revertible*, *specific*, and *unburdensome*; and introduce such theory to the design of social computing systems [1]
- **build consentful systems**—I embed affirmative consent into layers of social platforms, spanning functionalities, interfaces, and algorithms; e.g., a system that enables sensitive information sharing among those with less power, without revealing identities [2], consent-granting mechanisms for AI inferences [9]
- **rethink consent-granting interfaces and business models**—because tech companies obtain users' consent via interfaces that are heavily impacted by their business models; e.g., redesigning more findable ad settings [3], studying users' perception of different kinds of social media's business models

I have published 13 papers, of which 6 are first-authored. 9 are published in top venues in HCI and social computing, such as ACM CHI and CSCW, and 2 were accepted to top-tier privacy venues, such as Privacy Enhancing Technologies Symposium. My research has been recognized with two honorable mentions [1, 4], a Meta Research PhD Fellowship, a Barbour Scholarship, and an EECS Rising Star. My work is also included in course materials at the University of Washington, UIUC, and Stanford University.

1. Theorizing and Building Systems Grounded in Affirmative Consent

Theory of Affirmative Consent. What exactly constitutes an affirmative "yes," especially for the design of social computing systems? This was an open question at the time when I started my PhD. I first developed a theoretical framework of affirmative consent, drawing from feminist scholarship because it stresses the agency of the entity with less hard power. Feminist scholars have argued for affirmative consent ("yes means yes"), which emphasizes that a person must ask for approval before initiating an interaction, instead of passively assuming the person will agree to it. That is, it emphasizes that every individual has the agency to clearly signal whether they want an interaction to happen.

I focused on how such an agency-centric perspective offers valuable insights for the design of social computing systems. Specifically, in a CHI 2021 paper, I defined that affirmative consent is *voluntary*, *informed*, *revertible*, *specific*, and *unburdensome* [1], and theorized how it explains many disparate problems on social platforms. The five principles make it clear that for a range of problems, such as tech companies' data collection without users' knowledge, triggering content on social media, and online harassment, the fundamental issue lies in technology designs that do not prioritize users' consent. For example, while prior research has documented issues with content feeds, the framework shows that the known issues are affirmative consent violations. Users often do not concretely know which content and accounts will appear on the feed (*informed*), which can lead to traumatizing experiences. People also have nuanced needs around content types that vary across time (*revertible*).

More importantly, my theory serves as a basis for the design of **consentful systems**. For example, applying the *revertible* principle allows us to envision a profile page where users can easily reverse their decisions about the visibility of online traces on a large scale—similar to GitHub's revert feature; applying the *voluntary* principle to platforms' content feeds means asking what users want to see this week, instead of assuming.

Broadly, this work puts a stake in the ground in terms of rethinking what role consent plays in the field of computing. The field of privacy within computing has primarily looked at consent from a model called "notice and choice." It emphasizes providing individuals with information to make informed decisions ("notice") and certain controls for managing how one's data is collected and used ("choice"). However, privacy experts agree that this model has resulted in, for example, terms of service that users do not read. In contrast,

the affirmative consent framework shifts the discourse from legality (e.g., "Are companies meeting privacy regulations' requirements?") to meaningful user consent: "What do users think they are actually consenting? Do we really have their permission?"

That paper won an ACM CHI Honorable Mention, and it was also included in UW and Stanford's course materials (UW CSE 599, Stanford CS 278). Overall, the framework enables novel ways to address a rich set of problems related to safety, privacy, and power.

Building Consentful Systems. To validate how the theory of affirmative consent could be applied across domains, I also design and build consentful systems [2, 4, 7]. Most recently, I wanted to tackle a problem where focusing on consent would have real-world impact, and found a testbed—PhD students who could benefit from communicating with each other about advising challenges, but who are afraid to do so due to the topic's sensitivity. Interviews with PhD students showed that there is a need for an anonymous system that enables sharing information about navigating challenging advising relationships [2, 8].

Consent is deeply relevant to developing such systems because it is intertwined with trust—one consents to providing sensitive information only with people one trusts, and careful handling of consent builds trust. In particular, it is important to let users *voluntarily* make *specific* decisions in terms of initiating or accepting interactions (affirmative consent's *voluntary* and *specific* principles). Users might want to give consent to read a post only to some groups of peers, or give consent to revealing themselves only under certain conditions, etc.

Focusing on these issues, I developed a social media platform called Moa, to enable PhD students to communicate with each other through forms of very granular forms of consent [2]. Moa lets users set what I call a "**consent boundary**" per post or comment, without knowing other users' full identities. A consent boundary is a mechanism for users to demarcate precisely who can view their posts or comments in terms of a range of traits: social identity (e.g., race, gender), advising experiences (e.g., those who are micromanaged; those who do not receive enough feedback), advising status, etc. For example, a user can set one post to be visible only to students who experienced micromanagement, while showing another post only to women. (Moa also involves background moderation and strict identity verification.)

Compared to existing access controls, consent boundaries are fundamentally *user-centric*—their flexible dimensions related to social identity, lived experiences, etc. are not offered by existing access controls. Consent boundaries are also easily adjusted per post and comment, and not a blanket setting that applies to all posts/comments. (This is based on affirmative consent's *specific* principle.) Existing systems tend to offer very coarse categories (e.g., public or private) or individual blocking, with few options in between.

Deployment of Moa showed that consent boundaries incentivize users to share more sensitive information. This could be seen as paradoxical at first—because consent boundaries restrict what is shared. But, by increasing user trust in the system, consent boundaries actually encourage users to share more. In an evaluation of the system, 47 PhD students signed up to use Moa during one month. 22.6% of posters and commenters used various consent boundaries, who in interviews said they tuned their audience in a way that they believed would maximize empathy and relevance in responses. A few participants even indicated they wanted to meet other students in-person, and others seriously considered implementing suggestions they received on Moa. These findings imply that Moa brought some users to the threshold of taking action to resolve problems.

Figure 1: Moa (https://moa.systems) is a functioning social platform that incorporates a novel concept we call a *consent boundary*—it allows users to demarcate precisely who can view their posts/comments, based on social identity traits, lived experiences, etc., without knowing others' identities.

Beyond interpersonal interactions, I apply the affirmative consent framework to problems of companies' power over users. In one project, I investigated how nuanced consent-granting interfaces can help address privacy issues around platforms' algorithmic inferences. My collaborators and I built a recommendation system that suggests art-related opportunities (e.g., concerts, theaters) to understand how users make consent decisions regarding algorithmic inferences in realistic settings [9]. Unlike existing platforms, I ensured the treatment conditions had granular consent-granting interfaces regarding two types of information—1) data participants *voluntarily* provided to the system (i.e., participants said yes to submitting information) and 2) *inferences* drawn from that data. In experiments, participants were asked to indicate consent for both types of information on a 4-point scale: 1) no consent, 2) consent to use for arts recommendations, 3) consent to use for ads within the platform, and 4) consent to use for ads outside the platform. Each level included consent for previous levels.

Results showed that users who experienced my designs were more informed about the difference between data they voluntarily provide and inferences a system makes from that data. Compared to participants who did not learn about inferences being made, those who *were aware of inferences* gave restrictive privacy decisions for their initially voluntarily provided data. Furthermore, awareness of the most sensitive (e.g., citizenship status) and incorrect inferences made participants choose more restrictive consent decisions for the use of such inferences, and also for the use of their explicitly provided answers. Designs grounded in affirmative consent allow users to withhold consent for an organization to use the inferred information, instead of letting companies passively assume users give consent to using both the voluntarily provided data and inferences.

2. Rethinking Social Platforms' Interfaces & Business Models for Users' Consent

Even though companies are required by regulation to provide consent-granting settings for data, many users are often not aware of to what extent their data is used for companies' profit. This violates the *informed* principle of affirmative consent. In my research, I rethink both platforms' interfaces and business models to provide meaningful consent-granting mechanisms for users, while balancing companies' needs for profit.

In one project, I focused on the problem of many available ad controls on platforms being difficult to find [10]—despite the fact that they are the most common way for users to have some control over their data. This violates affirmative consent's *unburdensome* and *informed* principles. Thus, I researched how ad setting interfaces could be redesigned [3]. Then, I studied how findable ad settings impact users' perception of the settings (and the associated social media company), by building a Chrome extension that offered an alternative to Facebook's ad control interfaces. I conducted an online field experiment on Facebook where users in the treatment conditions were not told about the designs in advance. Results showed that our interfaces increased the findability of Facebook's ad controls by up to 41% compared to the baseline, and improved users' perceived usability of the ad controls. These findings show that companies could indeed make ad controls more usable and consentful. More importantly, our work shows the importance of testing concrete designs of privacy interfaces to inform regulation—this work was invited by the FTC to be presented at PrivacyCon 2024.

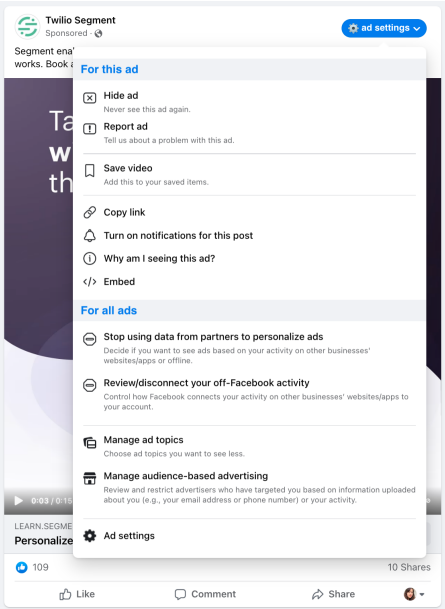

Figure 2: Example interface I designed for findable ad settings [3]. It provides direct links to privacy controls that originally require multiple clicks to be found when searching from Facebook's feed.

Beyond interfaces, I have also investigated alternative tech company business models and how they might interact with consent. Most social media companies rely on online behavioral advertising, even as regulators and academics have raised concerns about its harms. As a starting point to rethinking business models, I am leading a study on understanding how users perceive combinations of revenue models and privacy-enhancing technologies (PETs). I designed a large-scale survey that leverages conjoint analysis, an approach used by business scholars to understand how customers determine tradeoffs when given competing alternatives. Participants are asked to choose a platform after being shown a table with options—each row depicts a platform's attributes: 1. platform type (e.g., messenger, image-based), 2. type of organization running the platform (e.g., non-profit, private utility), 3. revenue model (e.g., advertising, freemium models, using platforms for free by providing data for their AI models), and 4. social media's usage of PETs (e.g., differential privacy). I am currently in the stage of pretesting the survey. To fund this work, I co-wrote a grant proposal with Prof. Florian Schaub (PI) that was awarded $75,000.

3. Research Vision

Ultimately, my goal is to reimagine forms of human-computer interactions where users are thoroughly respected, to support more effective and ethical computing. To realize this vision, I will first expand my research on consent to a wider range of problems, such as privacy issues with AI/ML development. Looking further ahead—for instance, in 10 years—I will also tackle challenges related to fairness and equity in systems. For example, I am increasingly interested in embedding plurality into AI/ML (e.g., incorporating different cultural perspectives) and designing mentoring systems for marginalized groups. These topics are closely tied to equity and fairness, and my research on consent provides a strong foundation for addressing them. This is because consent-granting behaviors are often intertwined with social identities, which are deeply connected to issues of equity and fairness [12]. Below, I describe two specific directions I will pursue in the next couple of years.

Embedding consent into AI/ML model development. Companies are developing AI/ML models based on massive amounts of data that they do not have explicit consent to use. While some attempts at policy have been made, such as the U.S. Executive Order on Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence, we are far from providing actionable guidelines to companies regarding model development. In terms of consent, technologists tend to assume binary forms of consent (opt-in/out).

I am especially interested in cases when AI/ML models (e.g., LLMs) use public social media posts as training data. Under the assumption that most users would not mind having their data used for some purposes, but would vehemently object to others, how should more granular forms of data-use consent be defined? For example, might users grant consent for training of grammar, but resist it for mental health content? Based on the affirmative consent principles [1], and leveraging my systems-building work [2, 3, 9], I will design novel consent-granting interfaces and systems, and then evaluate how users interact with them. For example, affirmative consent-granting mechanisms could 1) show users how their public social media posts could be used to develop AI/ML models, 2) let users make granular consent decisions, and 3) let users observe the model performance and revisit their decisions.

I would also like to understand how users' consent decisions might impact platforms' revenue. AI/ML models are used to power platforms' content feeds, and users' satisfaction with their content feeds likely impacts their engagement with ads or purchase of subscription services. This direction requires building systems, as well as leveraging methods from business management. The goal is to enable AI/ML development that can meet both users and companies' needs. I will obtain funding from NSF, Meta, MSR, and Google.

Building consentful systems for enabling concrete collective action. Many researchers have worked on online collective action, but prior research assumes that it happens among people who know they share a common cause, and who can readily find each other (e.g., [11]). I am interested in situations where people do not know if others share similar issues, and in any case cannot easily find allies. Moa (described above) has enabled PhD students to have risky discussions about advising relationships, but so far, it has stopped at discussion [2]. Building on it to enable collective action is my next goal.

One direction I will pursue is developing mechanisms for users to build trust to the point that they will reveal their identities to one another. (Some kinds of collective action will be facilitated when the actors know one another.) I will explore information escrows, which are systems that connect two parties, only if a mutual condition is met. However, traditional information escrows are binary—either a match happens or it does not; users cannot observe others until a condition is met. I hypothesize that a middle ground such as what Moa offers—where anonymous users labeled by non-identifying usernames can engage in conversations fully honoring consent—might create a gentler on-ramp for establishing the trust needed for an escrow condition to be met.

Another direction I will pursue is developing systems for users to arrive at concrete plans for collective action—even when individuals remain anonymous. There are multiple consent issues at play here. For example, for those who do not reveal their identity, what kind of information do they consent to sharing? How do users ask for others' consent in a feasible way, so that it does not impede collective action?

I will leverage Moa to longitudinally work on cycles of building, deploying, and evaluating interventions. I also plan to expand to other contexts, such as tech companies. For example, how do we design systems for potential whistleblowers to safely connect with each other to verify suspicions about companies' wrong-doing? I will obtain funding for this line of work from NSF based on my findings from Moa (e.g., NSF 24-604).

Conclusion. Centering users' consent to reimagine and develop technologies has never been more important. Consent problems are proliferating with the evolution of social media and the rise of AI/ML. In my research, I design, build, and deploy systems based on a rich theoretical understanding of consent. I envision a future where users have more agency over their data and interactions on the internet—a world where users have more of a voice in how their online peers and technology companies treat them.

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